

DATA ARCHITECTURE & DATASET SPECIFICATION

Trustworthy AI for Protein Engineering

Discovering Mutation-Space Reliability and Applying It to Industrial Carbon-Capture Enzyme Design

Version 2.0

Document Purpose

This document defines every dataset used throughout the project.

The purpose of this specification is to ensure:

Consistency

Reproducibility

Traceability

Validation

Schema Stability

Every dataset must have:

Defined Purpose

Defined Schema

Defined Inputs

Defined Outputs

Defined Dependencies

Defined Validation Rules

No dataset may exist outside this specification.

Data Philosophy

The project follows a simple principle:

Every Scientific Conclusion

must be traceable

to a specific dataset.

All datasets must be:

Immutable

Versioned

Auditable

Reproducible

Derived datasets must never overwrite source datasets.

Instead:

New Dataset

↓

New Version

↓

New File

must be created.

Master Data Flow

The project data architecture follows the sequence:

Megascale-D



Empirical Sparsity

Evolutionary Sparsity

Structural Sparsity



Combined Sparsity



Experiment 1

Antisymmetry



Experiment 2

Epistasis



Integrated Reliability Analysis



Reliability Score



TKT Mutation Landscape



Industrial Fitness Score



Candidate Ranking

SECTION 1

SOURCE DATASETS

Dataset S001

Megascale-D Mutation Dataset

Purpose:

Primary mutation benchmark dataset
used throughout discovery
and validation experiments.

File:

megascale_d.parquet

Primary Key:

mutation_id

Core Fields:

mutation_id

protein_id

position

wildtype

mutant

experimental_ddg

Used By:

Empirical Sparsity

Experiment 1

Experiment 2

Dataset S002

Protein Sequence Dataset

Purpose:

Input sequences for
ESM calculations.

File:

protein_sequences.parquet

Primary Key:

protein_id

Core Fields:

protein_id

sequence

sequence_length

Dataset S003

Protein Structure Dataset

Purpose:

Structural source
for SASA calculations
and active-site analysis.

File:

protein_structures.parquet
,

Primary Key:

protein_id

Core Fields:

protein_id

pdb_id

chain_id

structure_path

SECTION 2

SPARSITY DATASETS

Dataset D101

Empirical Sparsity

File:

empirical_sparsity.parquet

Purpose:

Store experimentally-derived
mutation density information.

Primary Key:

mutation_id

Schema:

mutation_id

mutation_frequency

normalized_frequency

empirical_sparsity

Valid Range:

empirical_sparsity

0.0 → 1.0

Dataset D102

Evolutionary Sparsity

File:

evolutionary_sparsity.parquet

Purpose:

Store ESM-derived
evolutionary rarity information.

Primary Key:

mutation_id

Schema:

mutation_id

esm_probability

log_probability

evolutionary_sparsity

Valid Range:

0.0 → 1.0

Dataset D103

Structural Sparsity

File:

structural_sparsity.parquet

Purpose:

Store SASA-derived
structural constraint information.

Primary Key:

mutation_id

Schema:

mutation_id

sasa

normalized_sasa

structural_sparsity

Valid Range:

0.0 → 1.0

Dataset D104

Combined Sparsity

File:

combined_sparsity.parquet

Purpose:

Unified sparsity framework
used throughout the project.

Primary Key:

mutation_id

Schema:

mutation_id

empirical_sparsity

evolutionary_sparsity

structural_sparsity

combined_sparsity

Valid Range:

0.0 → 1.0

This dataset serves as the primary explanatory variable for all scientific analyses.

SECTION 3

EXPERIMENT 1 DATASETS

Dataset D201

Antisymmetry Predictions

File:

antisymmetry_predictions.parquet

Purpose:

Store forward and reverse prediction results.

Schema:

mutation_id

forward_ddg

reverse_ddg

predictor

Dataset D202

Antisymmetry Results

File:

antisymmetry_results.parquet

Purpose:

Store thermodynamic consistency measurements.

Primary Key:

mutation_id

Schema:

mutation_id

forward_ddg

reverse_ddg

antisymmetry_error

combined_sparsity

Validation Rules:

antisymmetry_error ≥ 0

SECTION 4

EXPERIMENT 2 DATASETS

Dataset D301

Double Mutation Dataset

File:

double_mutants.parquet

Purpose:

Store experimentally
characterized mutation pairs.

Primary Key:

pair_id

Schema:

pair_id

mutation_a

mutation_b

experimental_ddg_ab

Dataset D302

Epistasis Results

File:

epistasis_results.parquet

Purpose:

Store epistasis calculations
and prediction errors.

Primary Key:

pair_id

Schema:

pair_id

experimental_epistasis

predicted_epistasis

epistasis_error

combined_sparsity

Validation Rules:

epistasis_error >= 0

SECTION 5

INTEGRATED RELIABILITY DATASETS

Dataset D401

Integrated Reliability Analysis

File:

integrated_reliability_analysis.parquet

Purpose:

Unified dataset supporting
the central scientific claim.

Primary Key:

record_id

Schema:

record_id

mutation_id

combined_sparsity

antisymmetry_error

epistasis_error

Scientific Purpose:

Evaluate

Sparsity



Antisymmetry



Epistasis

Dataset D402

Reliability Scores

File:

reliability_scores.parquet

Purpose:

Transform sparsity into
an engineering metric.

Primary Key:

mutation_id

Schema:

mutation_id

combined_sparsity

reliability_score

reliability_category

Allowed Categories:

High

Moderate

Low

Very Low

SECTION 6

TKT APPLICATION DATASETS

Dataset D501

TKT Mutation Landscape

File:

tkt_single_mutation_landscape.parquet

Purpose:

Store every possible
single mutation in TKT.

Primary Key:

mutation_id

Schema:

mutation_id

position

wildtype

mutant

mutation_label

Example:

A125V

Expected Size:

Protein Length

×

19 Substitutions

Dataset D502

TKT Mutation Analysis

File:

tkt_mutation_analysis.parquet

Purpose:

Master TKT engineering dataset.

Primary Key:

mutation_id

Schema:

mutation_id

evolutionary_sparsity

structural_sparsity

combined_sparsity

reliability_score

predicted_stability

Dataset D503

Active-Site Analysis

File:

active_site_analysis.parquet

Purpose:

Support control analysis.

Primary Key:

mutation_id

Schema:

mutation_id

distance_to_active_site

combined_sparsity

reliability_score

predicted_stability

Validation:

distance_to_active_site >= 0

SECTION 7

INDUSTRIAL FITNESS DATASETS

Dataset D601

Industrial Fitness Scores

File:

industrial_fitness_scores.parquet

Purpose:

Store engineering ranking metrics.

Primary Key:

mutation_id

Schema:

mutation_id

predicted_stability

reliability_score

evolutionary_plausibility

industrial_fitness_score

rank

Valid Range:

industrial_fitness_score

0.0 → 1.0

Dataset D602

Top Candidate Mutations

File:

top_candidate_mutations.csv

Purpose:

Final engineering output.

Schema:

rank

mutation

reliability_score

predicted_stability

industrial_fitness_score

Sorted:

Descending

Industrial Fitness Score

SECTION 8

FIGURE DATASETS

Dataset D701

Discovery Figure Dataset

Source:

combined_sparsity.parquet

antisymmetry_results.parquet

Primary Figure:

Combined Sparsity

vs

Antisymmetry Error

Dataset D702

Validation Figure Dataset

Source:

combined_sparsity.parquet

epistasis_results.parquet

Primary Figure:

Combined Sparsity

vs

Epistasis Error

Dataset D703

Integrated Reliability Dataset

Source:

integrated_reliability_analysis.parquet

Primary Figure:

Sparsity

↓

Antisymmetry

↓

Epistasis

Dataset D704

TKT Landscape Dataset

Source:

tkt_mutation_analysis.parquet

Primary Figure:

Reliability Landscape

Dataset D705

Industrial Fitness Dataset

Source:

industrial_fitness_scores.parquet

Primary Figure:

Top Ranked Mutations

DATASET LINEAGE

The complete dataset lineage is:

Megascale-D

↓

Empirical Sparsity

↓

Evolutionary Sparsity

↓

Structural Sparsity

↓

Combined Sparsity

↓

Antisymmetry Results

↓

Epistasis Results

↓

Integrated Reliability Analysis

↓

Reliability Scores

↓

TKT Mutation Landscape

↓

Industrial Fitness Scores

↓

Top Candidate Mutations

DATA VALIDATION REQUIREMENTS

Every dataset must pass:

Schema Validation

Type Validation

Range Validation

Primary Key Validation

Missing Value Validation

before downstream execution.

Any validation failure must stop execution immediately.

PROJECT MASTER DATA PRODUCTS

The final project must produce:

combined_sparsity.parquet

antisymmetry_results.parquet

epistasis_results.parquet

integrated_reliability_analysis.parquet

reliability_scores.parquet

tkl_single_mutation_landscape.parquet

tkl_mutation_analysis.parquet

active_site_analysis.parquet

industrial_fitness_scores.parquet

top_candidate_mutations.csv

These datasets collectively support all scientific conclusions, figures, publication outputs, science-fair materials, and industrial engineering recommendations generated by the project.

End Of Data Architecture & Dataset Specification